

Summary of Evaluation Metrics Used

1. Root Mean Squared Error (RMSE)

- Measures the difference between predicted and actual ratings.
- Lower RMSE indicates better prediction accuracy.

2. Precision@K

- Measures the proportion of top-K recommended items that are relevant.
- High precision means most recommendations are relevant.

3. Recall@K

- Measures the proportion of relevant items that are recommended in the top-K.
- High recall means the system captures a larger portion of relevant items.

Table of Model Results

Model	RMSE (Known)	Precision@5 (Known)	Recall@5 (Known)	RMSE (Cold- Start)	Precision@5 (Cold)	Recall@5 (Cold)
Collaborative Filtering	0.9236	0.4424	0.9315	0.9277	0.2095	1.0000
Content- Based Filtering	1.3426	0.4423	0.9314	1.3424	0.2095	1.0000

Comparison of Models Based on Performance

- Collaborative Filtering outperforms Content-Based Filtering significantly on:
 - RMSE (both known and cold-start) → indicating better rating predictions.
- Both models perform almost identically on:
 - Precision@5 and Recall@5, for both known and cold-start users.

- For Cold-Start scenarios, while recall remains perfect, precision drops sharply for both, reflecting challenges in relevance ranking despite full coverage.

Discussion on Performance

- **Collaborative Filtering** is the better overall model:
 - More accurate predictions (lower RMSE).
 - Slight edge in known-user recommendation precision.
- **Content-Based Filtering**, while similar in top-5 relevance metrics, **suffers from high RMSE**, indicating less reliable rating predictions.
- Both models handle **cold-start recall** well (likely due to setup), but **low precision** suggests that while all relevant items are shown, many irrelevant ones are too.